showed that the change in electric field also produces magnetic field which is

$$\oint_l \vec{B}\cdot\vec{dl} = -\frac{\partial}{\partial t}\Phi_E = -\frac{\partial}{\partial t}\oint_S \vec{B}\cdot\vec{dS} \qquad (5.2)$$

$$\underbrace{\oint_l \vec{E}\cdot\vec{dl}}_{\text{Magnetic field induced along a closed loop}} = \underbrace{-\frac{\partial}{\partial t}\Phi_E}_{\text{Variation of electric flux with time}} = \underbrace{-\frac{\partial}{\partial t}\oint_S \vec{B}\cdot\vec{dS}}_{\text{Changing electric flux } \Phi_E \text{ in the region enclosed by the loop}}$$

where Φ_B is the electric flux. This is known as Maxwell's law of induction, which explains that the magnetic field $\vec{B}$ induced along a closed loop by the changing electric flux Φ_E in the region encircled by that loop. This in turn, explains the existence of radio waves, gamma rays, infrared rays, etc.

In order to understand how the changing electric field produces magnetic field, let us consider a situation of 'charging a parallel plate capacitor' shown in Figure 5.3 Assume that the medium in between the capacitor plates is a non-conducting medium.

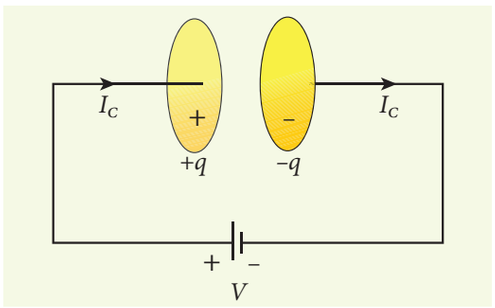


Figure 5.3 charging of capacitor

The electric current passing through the wire is the conduction current I_C. This current generates magnetic field around the wire (refer Unit 3) connected across the capacitor. Therefore, when a magnetic needle is kept near the wire, deflection is observed. In order to compute the strength of magnetic field at a point, we use Ampere's circuital law (from Unit 3) which states that 'the line integral of the magnetic field $\vec{B}$ around any closed loop is equal to μ_o times the net current I threading through the area enclosed by the loop'. Ampere's law in equation form is

$$\oint_l \vec{B}\cdot\vec{dl} = \mu_o I(t) \qquad (5.3)$$

where μ_o is the permeability of free space.

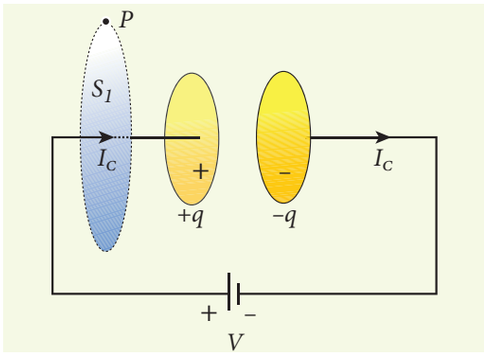


Figure 5.4 Applying Ampere's circuital law - loop enclosing surface

To calculate the magnetic field at a point P near the wire as shown in Figure 5.4, let us draw an amperian loop (circular loop) which encloses the surface S_1 (circular surface). Therefore, using Ampere's circuital law (equation 5.3), we get

$$\oint_{S_1} \vec{B}\cdot\vec{dl} = \mu_0 I_C \qquad (5.4)$$

where I_c is the conduction current.

Suppose the same loop is enclosed by balloon shaped surface S_2 as shown in Figure 5.5. This means that the boundaries of two surfaces S_1 and S_2 are same but shape of the enclosing surfaces are different (first surface (S_1) is circular in shape and second one is balloon shaped surface (S_2)). As the Ampere's law applied for a given closed loop does not depend on shape of the enclosing surface, the integrals will give the same answer. But by applying Ampere's circuital law (equation 5.3), we get

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